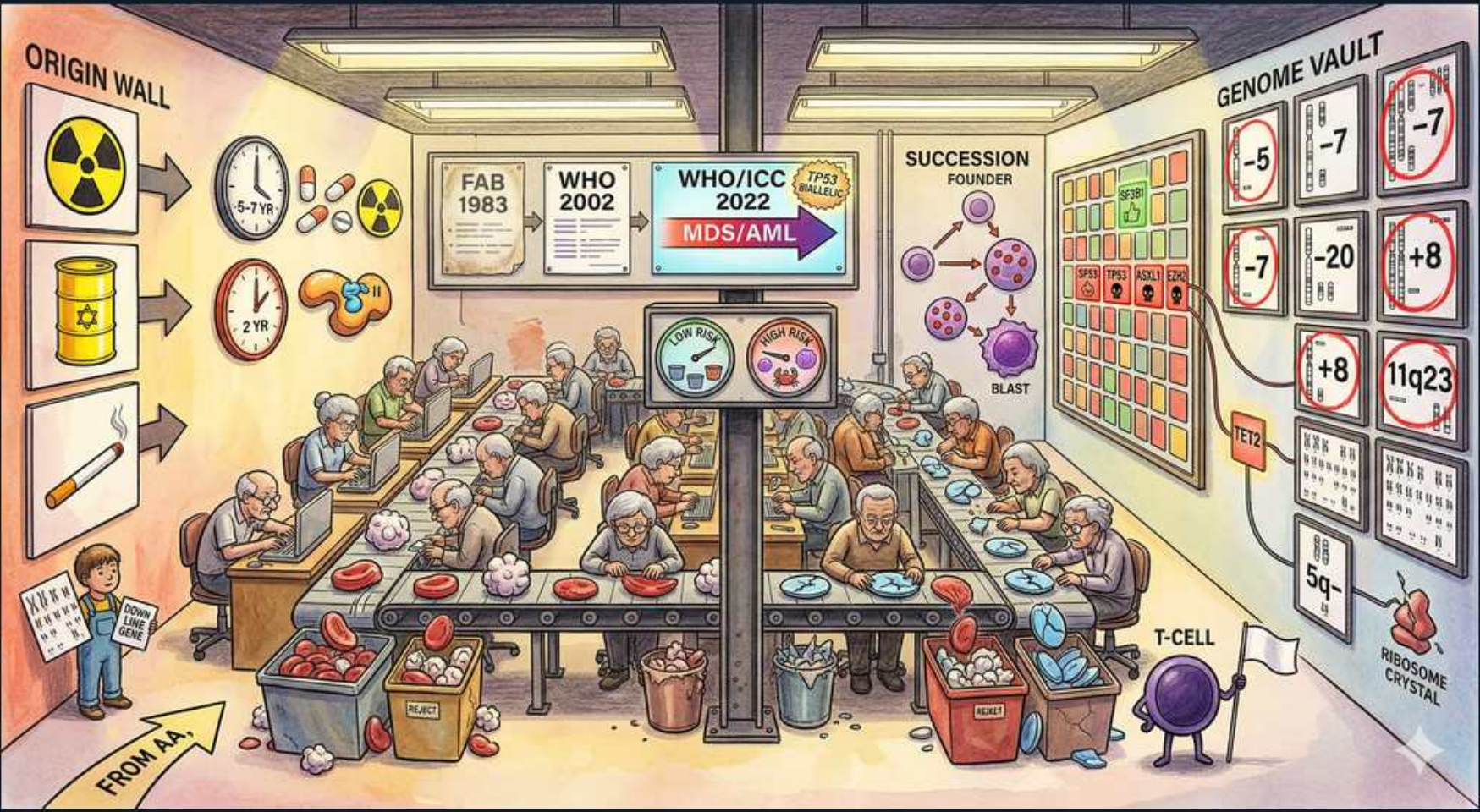


MDS — The Dysplastic Assembly Line, Classification & Cytogenetics Vault





1. Origin Wall — secondary MDS triggers

WHAT YOU SEE

Radiation symbol, chemo pills, benzene barrel and a cigarette on the left wall.

WHAT IT ENCODES

Therapy-related/secondary MDS arises from DNA-damaging exposures: prior alkylating chemotherapy (cyclophosphamide, melphalan, busulfan), ionizing radiation, benzene/solvents, and tobacco smoke. These cause acquired clonal mutations in a hematopoietic stem cell that then expands. Secondary MDS accounts for ~10–15% of cases and is enriched for TP53 mutations and complex karyotypes. Mnemonic: 'RAB-S' — Radiation, Alkylators, Benzene, Smoking.

BOARD TRAP

A patient cured of breast cancer or lymphoma 5–7 years ago now has pancytopenia. WRONG answer: 'anemia of chronic disease' or relapsed lymphoma. Discriminator: therapy-related MDS carries a WORSE prognosis than de novo MDS, is driven by –5/–7/TP53, and is largely chemo-refractory — it is NOT managed identically to primary MDS and often warrants early transplant referral.

2. Latency clocks (5–7 yr vs ~2 yr)

WHAT YOU SEE

Two clocks marked ~6–7 yr and ~2 yr.

WHAT IT ENCODES

The two classic t-MDS pathways differ by latency and genetics. Alkylating agents → MDS at ~5–7 years, typically with deletions/monosomies of chromosomes 5 and 7 (–5/del(5q), –7), and usually pass through an MDS phase before AML. Topoisomerase-II inhibitors (etoposide, anthracyclines) → ~1–3 years with balanced translocations of 11q23 (KMT2A/MLL) or 21q22 (RUNX1-RUNX1T1), often presenting directly as AML without an MDS prodrome.

BOARD TRAP

Stem asks: leukemia 18 months after etoposide for testicular cancer, t(9;11). WRONG answer: alkylator-induced –7 MDS. Discriminator: SHORT latency (1–3 yr) + balanced 11q23/MLL translocation = topoisomerase-II inhibitor, not the longer-latency alkylator pattern with –5/–7.

3. ‘FROM AA’ — clonal evolution

WHAT YOU SEE

An arrow labelled ‘FROM AA’ entering the room.

WHAT IT ENCODES

MDS can evolve from acquired aplastic anemia or PNH as an immune-privileged clone (often with del(7) or PIGA/other mutations) escapes the autoimmune attack and expands — ‘clonal evolution.’ Roughly 10–15% of AA patients develop clonal disease (MDS/AML) over years, and PNH clones are detectable in a subset. Long-term monitoring with CBC and periodic marrow/cytogenetics is therefore standard after AA treatment.

BOARD TRAP

A young AA patient in remission on cyclosporine develops new dysplasia and monosomy 7. WRONG answer: relapsed aplastic anemia or drug effect. Discriminator: emerging dysplasia + clonal cytogenetics (especially –7) in a treated AA/PNH patient = clonal evolution to MDS, which changes management toward transplant rather than more immunosuppression.

4. Classification timeline (FAB → WHO/ICC 2022)

WHAT YOU SEE

Posters: FAB 1983 → WHO 2002 → WHO/ICC 2022 MDS/AML.

WHAT IT ENCODES

MDS classification has shifted from pure morphology toward genetics. The 1983 FAB system used blast % alone (RA, RARS, RAEB, RAEB-t, CMML). WHO 2002/2016 added lineage dysplasia, ring-sideroblast count, and isolated del(5q). The 2022 WHO/ICC systems define genetically-defined entities — MDS with low blasts and SF3B1 mutation, MDS with biallelic TP53 — and lowered/abolished the rigid 20% blast wall for certain genetic AMLs. Trend mnemonic: 'morphology → mutations.'

BOARD TRAP

A vignette labels disease 'RAEB-t' (refractory anemia with excess blasts in transformation, 20–30% blasts). WRONG answer: treat as high-risk MDS. Discriminator: RAEB-t was abolished — ≥20% blasts is now classified and treated as AML, so the old FAB category should be reclassified, not perpetuated.

5. Blast threshold MDS vs AML

WHAT YOU SEE

The central 'MDS/AML' divider arrow with a single round BLAST cell on it — the blast acts as the gatekeeper marking the <20% (MDS) versus >=20% (AML) boundary.

WHAT IT ENCODES

The historical dividing line is blast percentage: MDS has <20% myeloblasts in blood or marrow, AML has ≥20%. MDS subtypes are further split by blast count (e.g., MDS-IB1 = 5–9% marrow blasts, MDS-IB2 = 10–19%). However, AML-defining recurrent genetic abnormalities override the count: t(8;21), inv(16)/t(16;16), t(15;17)/PML-RARA, and KMT2A rearrangements diagnose AML even when blasts are below 20%.

BOARD TRAP

Marrow shows 12% blasts with inv(16). WRONG answer: MDS with excess blasts (MDS-IB2). Discriminator: core-binding-factor and APL genetics define AML at ANY blast count — the 20% cutoff is not absolute, so this is AML requiring induction, not MDS supportive care.

6. TP53 multi-hit

WHAT YOU SEE

A red alarm tag reading 'TP53 multi-hit' pinned beside the classification posters — the red colour flags this biallelic-TP53 entity as the worst-prognosis, chemo-resistant genetic group.

WHAT IT ENCODES

Biallelic ('multi-hit') TP53 — two TP53 mutations, or one mutation plus 17p deletion/LOH — defines a distinct, very-poor-risk WHO/ICC entity. It is strongly associated with complex karyotype, therapy-related disease, and rapid AML progression with median survival often under a year. It is profoundly resistant to conventional chemotherapy, including hypomethylating agents, and even transplant outcomes are poor.

BOARD TRAP

Older patient with complex-karyotype MDS and biallelic TP53; the team plans standard '7+3' induction expecting cure. WRONG answer: intensive chemo will achieve durable remission. Discriminator: TP53 multi-hit disease responds poorly to standard chemo — the correct move is clinical trial enrollment and/or transplant discussion, with realistic prognostic counseling, not routine induction.



7. Dysplastic assembly line (elderly clone)

WHAT YOU SEE

Rows of elderly workers building defective cells on a conveyor.

WHAT IT ENCODES

MDS is a clonal stem-cell disorder of older adults (median age ~70). A mutated stem cell drives ineffective hematopoiesis; progenitors proliferate but die by intramedullary apoptosis, so the marrow is usually HYPERcellular yet the periphery is cytopenic. The most common founding mutations hit RNA splicing (SF3B1, SRSF2, U2AF1), epigenetics (TET2, ASXL1, DNMT3A), and signaling — the same clones seen in clonal hematopoiesis (CHIP) that can progress to MDS.

BOARD TRAP

Elderly patient with pancytopenia; marrow is hypercellular. WRONG answer: aplastic anemia. Discriminator: cytopenias with a CELLULAR (not empty/fatty) marrow plus dysplasia = MDS via ineffective hematopoiesis; aplastic anemia shows a HYPOcellular, fatty marrow without dysplasia.

8. Defective products — tri-lineage dysplasia

WHAT YOU SEE

Bins of malformed red cells, white cells and platelets.

WHAT IT ENCODES

Dysplasia spans the three myeloid lineages: erythroid (megaloblastoid/nuclear-budding RBCs, ring sideroblasts), granulocytic (hypogranular, hyposegmented pseudo-Pelger-Huët neutrophils), and megakaryocytic (small mono-/hypolobated micromegakaryocytes). This produces peripheral cytopenias — anemia (macrocytic, low retic) is most common, followed by neutropenia and thrombocytopenia. Diagnosis requires dysplasia in ≥10% of cells in at least one lineage.

BOARD TRAP

Pancytopenia with oval macrocytes and hypersegmented neutrophils. WRONG answer: jump to MDS and order a marrow. Discriminator: hypersegmentation and macro-ovalocytes also define B12/folate deficiency — you must exclude reversible mimics (B12, folate, copper, alcohol, drugs) BEFORE diagnosing MDS, because they cause identical megaloblastoid dysplasia.

9. Genome Vault — cytogenetics drive prognosis

WHAT YOU SEE

A wall of karyotype cards (the ‘Genome Vault’).

WHAT IT ENCODES

Cytogenetics are a cornerstone of risk: the revised IPSS-R sorts karyotypes into five groups (very-good, good, intermediate, poor, very-poor) that independently predict overall survival and time to AML. Conventional karyotyping (≥20 metaphases) ± FISH is mandatory at diagnosis; ~50% of de novo MDS has a clonal abnormality. The newer IPSS-M layers molecular mutations on top of this.

BOARD TRAP

Two patients have identical 8% blasts and similar counts but different karyotypes. WRONG answer: treat both the same based on blast count. Discriminator: IPSS-R RISK — not blasts or symptoms alone — drives therapy; cytogenetics can move one patient to low-risk supportive care and the other to high-risk HMA/transplant.

10. Poor-risk karyotypes (–7, –5, complex)

WHAT YOU SEE

Red cards: –7, –5, +8, 11q23, complex.

WHAT IT ENCODES

High-/very-high-risk cytogenetics include monosomy 7/del(7q), –5 (when part of a complex pattern), and complex karyotype (≥3 abnormalities) — the latter is very-poor risk and frequently TP53-associated. Trisomy 8 and 11q23 are intermediate/adverse. These predict shorter survival and faster AML transformation and push toward hypomethylating agents and early transplant evaluation.

BOARD TRAP

Vignette emphasizes ‘a chromosome-5 abnormality is favorable, so reassure.’ WRONG answer: assume any chr-5 lesion is good. Discriminator: ISOLATED del(5q) is favorable, but monosomy 7 and –5 within a COMPLEX karyotype are high-risk — don’t generalize the del(5q) good prognosis to monosomy 7 or complex disease.

11. Isolated del(5q) — favorable

WHAT YOU SEE

A ‘5q–’ card set apart at the bottom.

WHAT IT ENCODES

5q– syndrome: classically an older woman with macrocytic, transfusion-dependent anemia, NORMAL or HIGH platelet count, and hypolobated micromegakaryocytes. Isolated del(5q) (or with one additional non-chr7 abnormality) is a GOOD-prognosis WHO entity. Loss of RPS14 (ribosomopathy) drives anemia; lenalidomide (10 mg/day, 21 of 28 days) yields transfusion independence in ~65% and cytogenetic responses by suppressing the del(5q) clone.

BOARD TRAP

Low-risk transfusion-dependent anemia with isolated del(5q); team starts azacitidine. WRONG answer: hypomethylating agent first-line. Discriminator: del(5q) syndrome is THE MDS subtype treated with lenalidomide (after ESA failure or if EPO unlikely to work) — HMAs are reserved for higher-risk disease, not low-risk isolated del(5q).

12. T-cell / hypoplastic MDS

WHAT YOU SEE

A lone T-CELL figure by the vault.

WHAT IT ENCODES

Hypocellular (hypoplastic) MDS is a <10–20% subset with a HYPOcellular marrow that overlaps clinically with aplastic anemia. It is often associated with a cytotoxic T-cell/autoimmune mechanism, PNH clones, and HLA-DR15, and a meaningful fraction respond to immunosuppressive therapy (ATG ± cyclosporine), especially younger patients with low blasts and few high-risk cytogenetics.

BOARD TRAP

Young patient, hypocellular marrow, mild dysplasia — labeled straightforward aplastic anemia. WRONG answer: ignore clonality and treat purely as AA. Discriminator: hypoplastic MDS is distinguished from AA by definite morphologic dysplasia and clonal cytogenetics/mutations; finding these reclassifies it as MDS, though immunosuppression may still be appropriate for the hypocellular, low-risk phenotype.